

Blanka Balogh

Scientific Machine Learning Researcher | Toulouse, France
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SUMMARY

Scientific ML researcher specializing in AI-accelerated simulation and surrogate modeling of large-scale physics solvers. 7+ years building hybrid physics/ML systems: designing simulation campaigns to generate training data, curating simulation and observation datasets, training and evaluating neural network surrogates, and deploying them online in a large-scale operational Fortran solver on heterogeneous CPU/GPU HPC, and in leading research end-to-end, from hypothesis formulation to operational testing and publication.

TECHNICAL SKILLS AND METHODS

Deep Learning Frameworks: PyTorch, Lightning, Tensorflow, MLFlow, TensorBoard, anemoi

Scientific Computing and HPC: MPI, openMP, Slurm, Linux, Shell, Fortran, Python, (py)OASIS, git

Research Methods: surrogate modeling, hybrid physics/AI, probabilistic modeling, multifidelity, scientific ML, scientific computing, simulation engineering

Deep Learning methods: Neural Networks, Graph Neural Networks, Graph Transformers, Flow Matching

EXPERIENCE

Research Scientist

Oct 2022 – present

Météo-France, Centre National de Recherches Météorologiques

Toulouse, France

- Developed a graph transformer-based neural ocean emulator (PyTorch, anemoi) to replace a costly ocean model as the lower boundary forcing of a global atmospheric model using observations, enabling coupled-mode simulations that were previously computationally prohibitive.
- Developed hybrid physics/ML methods for atmosphere model ARP-GEM (50-100 km globally) using OASIS and PhyDLL: neural subgrid parameterizations, graph NN surface ocean model.
- Designed simulation campaigns using a large physics solver (10+ million lines of Fortran) to produce the high-fidelity data used to train the neural network surrogates.
- Extended a large-scale operational Fortran solver to support heterogeneous CPU/GPU HPC configurations, enabling faster NN inference online
- Deployed and evaluated online neural network components (Graph NNs, Flow Matching models) in a large-scale operational Fortran climate model
- Mentored interns and advised ML components of interdisciplinary research projects

PI of project GAIAC – French National Program INSU-SUN

Jan 2026 – Dec 2028

Subject: ML parameterizations with improved out-of-sample generalizability

Visiting scholar

Sept 2023 – Dec 2023

Pierre Gentine's group, Columbia University

New York, NY, USA

EDUCATION

Institut National Polytechnique de Toulouse

Sept 2019 – Oct 2022

PhD in Earth and Planetary sciences. Subject: hybrid AI/physics modeling in climate

Toulouse, France

Ecole Nationale de la Météorologie

Sept 2016 – Aug 2019

M.Sc.Eng. in Weather Forecasting and Numerical Modeling

Toulouse, France

INSA Toulouse

Sept 2018 – Aug 2019

M.Sc. exchange year in Applied Mathematics with a major in machine learning

Toulouse, France

SELECTED PUBLICATIONS

- **Balogh, B.**, Saint-Martin, D., Geoffroy, O., 2026: ARPGEM-ANEMONE: a physical atmosphere model coupled to a Graph Neural Network-based surface ocean emulator. Accepted at NeurIPS workshop 'Sim2Science' (Paris, Dec. 2026).
- **Balogh, B.**, Bhourri, M. A., Peng, L., Pritchard, M., Gentine, P., 2026: Probabilistic multi-fidelity climate model parameterization for better generalization and extrapolation. IOP Machine Learning: Earth., 10.1088/3049-4753/aea999.
- Germain, H., **Balogh, B.**, Geoffroy, O., and Saint-Martin, D., 2026: Improvement of a neural network convection scheme by including triggering and evaluation in present and future climates. Artif. Intell. Earth Syst., e250108, 10.1175/AIES-D-25-0108.1.
- **Balogh, B.**, D. Saint-Martin, and O. Geoffroy, 2025: Online Test of a Neural Network Deep Convection Parameterization in ARP-GEM1. Artif. Intell. Earth Syst., e240100, 10.1175/AIES-D-24-0100.1.
- **Balogh, B.**, Saint-Martin, D., and Ribes, A., 2022: How to calibrate a dynamical system with neural network based physics? Geophysical Research Letters, e2022GL097872, 10.1029/2022GL097872.
- **Balogh, B.**, Saint-Martin, D., and Ribes, A., 2021: A toy model to investigate stability of AI-based dynamical systems. Geophysical Research Letters, e2020GL092133, 10.1029/2020GL092133.

SELECTED PRESENTATIONS

- Invited Speaker: CNRS annual conference 'Défis Théoriques pour la science du climat', Paris, France (2026).
- Oral presentation: EGU, Vienna, Austria (2026, 2024).
- Poster presentation: Exclaim! Symposium, Zürich, Switzerland (2025).

SCIENTIFIC SERVICE AND LEADERSHIP

Associate Editor <i>Artificial Intelligence for the Earth Systems (American Meteorological Society)</i>	2023 – present
Co-convener at EGU <i>Session: Machine Learning for climate science, 50+ oral presentations and posters.</i>	2026-2027
External Reviewer <i>External Reviewer for a French National Research Agency (ANR) project</i>	2026
Scientific Program committee member for ICML <i>Reviewer for the workshop 'Machine Learning for Earth System Modeling'</i>	2024

RESEARCH PARTNERSHIPS

- Probabilistic multi-fidelity parameterization for subgrid-scale climate processes in collaboration with Columbia University (P. Gentine's group).
- Machine learning approaches for global wildfire burned areas prediction with CNRM/CNRS collaborators.
- Contributed to the development and debugging of open-source coupling tools for hybrid physics/ML climate workflows (PhyDLL, CERFACS collaboration)

RESEARCH SUPERVISION

- ML advisor for PhD projects on downscaling, urban climate modeling and wildfire prediction from observations
- Supervised M.Sc. and research internship projects on stochastic and physically consistent climate modeling using neural networks